

Input Form (IF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: ARABICMMLU | **Context:** MENA Multi-Country Arabic Tutoring System — Eight-Country Curriculum Deployment

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Input form aligns well with the deployment. All 14,575 questions are MSA [\[Q2\]](#), matching the formal medium of instruction across all eight target countries [\[Q29\]](#). The text-only, MCQ format with embedded contextual passages [\[Q37, Q38\]](#) is exactly the deployment's primary use case. Arabic-script fidelity is empirically confirmed [\[DATASET-D24, D25\]](#). The MSA-only scope appropriately excludes Darija (which the deployment explicitly excludes) and dialectal Arabic [\[Q92\]](#). The main form-level gap is French-medium Moroccan university content [\[Q92\]](#), which the deployment user explicitly identified as in-scope for Moroccan university science programs [elicitation A4]. A secondary form concern is OCR corruption observed in Management and Political Science (University) questions from Egyptian PDF sources [\[DATASET-D14, D15, D29\]](#), which compromises the input signal quality for those configurations. A minor anomaly is one Arabic Grammar question with English-language stem [\[DATASET-D19\]](#).

ID	Evidence
Q2	"Our data comprises 40 tasks and 14,575 multiple-choice questions in Modern Standard Arabic (MSA) and is carefully constructed by collaborating with native speakers in the region." (p.1)
Q29	"In public schools, Arabic is commonly used for teaching and assessment, while in international schools, English is the predominant language of instruction for most subjects, following either the UK or USA curriculum." (p.3)
Q37	"During manual data scraping, workers were instructed to include only questions accompanied by an answer key, and to discard questions containing multi-modal information (e.g., images, videos, or tables)." (p.4)
Q38	"If a question had additional contextual information (e.g., a passage referenced by several questions), the context was required to be included with each corresponding question." (p.4)
Q92	"The dataset primarily focuses on Modern Standard Arabic (MSA). However, multilingual and Arabic LLMs are often exposed to both MSA and dialectal Arabic." (p.9)
D14	Severely OCR-distorted text about authority distribution in organizations, from Egyptian source إلى ري ت ذت...ة ف ت خ س ل ا ت ي ر ا د ل ا ا ت ا ي ه ت ت د س ل ا و ص ا خ ت ل ا ل ا ن ي ت ب ة ط ل ت د ل ا ع ت ي ي ه ت ة ت ج ر د
D15	Severely OCR-corrupted management question with unreadable answer options ت ح ل ا ... ت م ح ر خ ر ل ا ... م ا د خ ت س ل ا ا ت س ن ا د ط ط خ ل ا ي ل ع ل ا ا ث م د ع ي ل ا
D19	In the following Quranic verse, what is the correct parsing of the word كَد — English-language question stem in an Arabic grammar dataset
D24	Calvin cycle question — ا ه ت ي ب ث ت م ت ي ت ل ا (CO2) ت ا ئ ي ز ج د د ع م ك ف ن ف ل ا ك ة ق ل ح ن ع ز و ك و ل غ ت ا ئ ي ز ج (3) ج ت ن ا ذ ا Palestine-sourced biology; universal factual content
D25	"The main goal of the 1928 Jordanian-British treaty was liberation from Mandate restrictions" — Jordanian national history ب ا د ت ن ا ل ا ك ف د و ي ق ن م د ا ل ب ل ر ر ح ت : ١٩٢٨ م ا ع ه ي ن ا ط ي ر ب ل ا ه ي ن د ر ا ل ا ه د ا ع م ل ل س ي ئ ل ا ف د ه ل ا ن ا ك
D29	Severely OCR-distorted Arabic text about conditions for an Islamic ruler — from Egyptian source م ل ا ر ر س ل ا ر ر ل و د ل ا ر ر ف م ك ا ر ر ل ل ا ر ر ف ا ر ر ف ا و ت ب ر ر ر ر ت ل ا ر ر س ل ا ل ا ط و ر ر ر ش ل ا ن ر ر م

Strengths

- MSA throughout matches the formal medium of instruction in all eight countries [\[Q2, Q29\]](#)
- MCQ + 2–5 options format directly matches deployment's primary use case [\[Q27\]](#)
- Text-only modality matches the deployment's text-based interface [\[Q93, infrastructure section\]](#)

- Contextual passages bundled with dependent questions [Q38] preserves single-shot evaluability

ID	Evidence
Q2	"Our data comprises 40 tasks and 14,575 multiple-choice questions in Modern Standard Arabic (MSA) and is carefully constructed by collaborating with native speakers in the region." (p.1)
Q27	"Each question has 2–5 candidate answers, with one correct answer." (p.3)
Q29	"In public schools, Arabic is commonly used for teaching and assessment, while in international schools, English is the predominant language of instruction for most subjects, following either the UK or USA curriculum." (p.3)
Q38	"If a question had additional contextual information (e.g., a passage referenced by several questions), the context was required to be included with each corresponding question." (p.4)
Q93	"ArabicMMLU is focused solely on text-based assessment, and the exploration of multimodal questions is left for future work." (p.9)

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Signal distributions align: MSA Arabic text input matches the deployment's text-based interface and matches the formal language of instruction across all eight countries [Q2, Q29]. No image/audio mismatch since both the benchmark and deployment exclude multimodal content [Q37, Q93].

IF-2: Regional infrastructure supports text-based delivery: ITU 2024 data shows ~100% internet penetration in UAE/KSA/Kuwait [WEB-8], ~83-88% in Levant per Arab Barometer [WEB-10], and majority mobile-dominant access across all eight [WEB-11]. Mobile-dominant access is compatible with text MCQ delivery.

IF-3: Domain-specific form gap: French-medium Moroccan university science content is entirely outside benchmark scope [Q92]; deployment explicitly identifies this as in-scope for Moroccan universities [elicitation A4]. No companion benchmark identified [gap_id 6]. Additionally, OCR corruption in some Egyptian-sourced university PDFs [DATASET-D14, D15, D29] degrades input form for those subjects.

IF-4: Documented form mismatches: (a) French-medium Moroccan university content absent [Q92, gap_id 6]; (b) OCR-corrupted text in Management and Political Science university questions [DATASET-D14, D15, D29]; (c) one English-stem grammar question [DATASET-D19]; (d) several configurations have null country/level metadata limiting form-level diagnostics [DATASET-D26, D27].

ID	Evidence
Q2	"Our data comprises 40 tasks and 14,575 multiple-choice questions in Modern Standard Arabic (MSA) and is carefully constructed by collaborating with native speakers in the region." (p.1)
Q29	"In public schools, Arabic is commonly used for teaching and assessment, while in international schools, English is the predominant language of instruction for most subjects, following either the UK or USA curriculum." (p.3)
Q37	"During manual data scraping, workers were instructed to include only questions accompanied by an answer key, and to discard questions containing multi-modal information (e.g., images, videos, or tables)." (p.4)
Q92	"The dataset primarily focuses on Modern Standard Arabic (MSA). However, multilingual and Arabic LLMs are often exposed to both MSA and dialectal Arabic." (p.9)
Q93	"ArabicMMLU is focused solely on text-based assessment, and the exploration of multimodal questions is left for future work." (p.9)
D14	على رأي تذا...ة فلت خسل اة تي رادل اة اتي هت تدسل او صا ختل ال اة ني تب عطل تدل اة تي هت عة جرد — Severely OCR-distorted text about authority distribution in organizations, from Egyptian source

